

Common Errors And Solutions

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Section 1: multi_brains_pre Package

1.1 Configuration File Related Errors

Error 1.1.1: Configuration File Not Found

Error Log:

```
multi_brains_pre_setting.yaml file not found, Please check: whether the file exists;: {config_file}
```

Cause:

- The multi_brains_pre_setting.yaml configuration file path does not exist

- The file was deleted or moved

Solution:

1. Copy multi_brains_pre_setting.yaml from the attached source code to the specified directory
-

Error 1.1.2: YAML Parsing Failed**Error Log:**

```
multi_brains_pre_setting.yaml file YAML parsing failed: {exc}, Please check:  
whether the file format has been modified
```

Cause:

- YAML file format error (incorrect indentation, special characters not escaped, etc.)

Solution:

1. Modify multi_brains_pre_setting.yaml, refer to the factory file format in the attached source code
-

1.2 ASR Voice Recognition Related Errors**Error 1.2.1: ASR Engine Initialization Failed****Error Log:**

```
ASR Engine: {model_name} initialized failed
```

Cause:

- API Key configuration error or invalid
- Model name not supported
- Network issue (online ASR)
- Key file decryption failed (iFlytek ASR)

Solution:

1. Check if `ALIYUN_API_KEY` is correctly configured
 2. Confirm the model name being used is in the supported list:
 - paraformer-realtime-v2/v1
 - paraformer-realtime-8k-v2/v1
 - gummy-realtime-v1/gummy-chat-v1
 3. For iFlytek ASR, check if the key file path and decryption are correct
 4. Test network connection
-

Error 1.2.2: Voice Recognition Failed**Error Log:**

```
asr recognize failure  
online ASR recognition failed: {error_message}
```

Cause:

- API invalid
- Online voice recognition service unavailable

Solution:

1. Check network connection stability
 2. Check if API quota is exhausted
-

Error 1.2.3: VAD Engine Initialization Failed

Error Log:

```
VadEngine initialization failed  
VadEngine initialization exception: {exc}
```

Cause:

- VAD model file does not exist
- Model path configuration error
- voice_toolbox library not installed correctly

Solution:

1. Check if the `silero_vad.int8.onnx` model file was deleted
 2. Download the VAD model file from the attached source code
-

Error 1.2.4: Audio Stream Open Failed

Error Log:

```
Audio stream open failed: {exc}
```

Cause:

- Microphone device not connected
- Audio device occupied by other programs

Solution:

1. Check if the microphone device is properly connected
 2. Use `arecord -l` to view available audio devices
 3. Close other programs occupying the audio device
-

Error 1.2.5: No Speech Detected

Warning Log:

```
No speech detected, timeout exit.
```

Cause:

- Microphone volume too low
- Environment noise too loud

Solution:

1. Increase microphone input volume
 2. Reduce environment noise
 3. Speak closer to the microphone
-

Error 1.2.6: Unsupported Language**Warning Log:**

```
Unsupported language: {language}, only 'zh' and 'en' are supported
```

Cause:

- Configured unsupported language code
- LANGUAGE parameter set incorrectly

Solution:

1. Change the LANGUAGE parameter to 'zh' (Chinese) or 'en' (English)
 2. Check the language setting in the configuration file
`$HOME/M3Pro_ws/src/m3pro_bringup/config/multi_brains_pre_setting.yaml`
-

1.3 TTS Voice Synthesis Related Errors**Error 1.3.1: TTS Engine Initialization Failed****Error Log:**

```
Failed to initialize Piper TTS engine: {e}  
Failed to initialize Baidu TTS engine: {e}  
Tongyi TTS_Engine initialized failed  
Baidu TTS_Engine initialized failed  
Xunfei TTS_Engine initialized failed  
Piper TTS_Engine initialized failed
```

Cause:

- TTS model file missing
- API Key configuration error
- Configuration file read failed
- Key file decryption failed (iFlytek TTS)

Solution:**1. Piper TTS:**

- Check if the model file exists in the `~/MODELS/tts/` directory
- Confirm both `.onnx` and `.onnx.json` files exist

2. Online TTS (Aliyun/Baidu/iFlytek):

- Check if the corresponding API Key is correctly configured
- Verify network connection

3. Check if the configuration file format is correct

Error 1.3.2: Voice Synthesis Failed

Error Log:

```
PiperTTS Failed to synthesize text: {e}  
Speech synthesis failed. Check whether the TTS model is available
```

Cause:

- TTS engine not correctly initialized?
- Online voice service unavailable?

Solution:

1. Confirm TTS engine is correctly initialized
 2. View specific error messages for targeted handling
-

1.4 Voice Wake-up Module Serial Communication Related Errors

Error 1.4.1: Serial Port Open Failed

Error Log:

```
[kws_mic] Failed to open serial {port}: {e}  
Cannot open serial port {port}, please check voice wakeup module connection
```

Cause:

- Voice wake-up serial device does not exist
- Hardware connection issue

Solution:

1. Check if the serial device exists: `ls -l /dev/mic`
 2. Check if the hardware connection is secure
-

1.4.2: Serial Reconnection Failed

Warning Log:

```
[kws_mic] Max reconnect attempts ({max_attempts}) reached, giving up  
[kws_mic] Reconnect attempt {attempt} failed: {e}
```

Cause:

- Hardware device disconnected

Solution:

1. Re-plug the USB device
-

1.4.3: Serial Send Failed

Error Log:

```
send fail: {e}
```

Cause:

- Serial connection already disconnected
- Buffer overflow
- Voice wake-up device not responding

Solution:

1. Check serial connection status
 2. Restart the program
 3. Check if the hardware device is working normally
-

1.5 ROS Communication Related Errors

Error 1.5.1: Action Server Unavailable

Warning Log:

```
play_audio action server not available, skip wakeup voice  
play_audio action server not available, skip error audio  
play_audio action server not available, skip voice
```

Cause:

- common_service node not started

Solution:

1. Confirm the common_service node is started
 2. Use `ros2 action list` to check available actions
-

Error 1.5.2: Action Goal Rejected

Warning Log:

```
play_audio goal rejected  
TTS play goal rejected
```

Cause:

- Audio file path does not exist

Solution:

1. Check if the audio file exists
 2. View specific rejection reason logs
-

1.6 LLM Service Related Errors

Error 1.6.1: Dify Connection Failed

Error Log:

```
Dify LLM connection failed, Please check whether DIY dify has been started, API
and BASE_URL settings
Error log: {error_message}
Dify service connection failed: {status_code}
Dify service connection failed: {str(e)}
```

Cause:

- Dify service not started
- BASE_URL configuration error
- **API Key invalid** (highest probability, dify application's API does not match DIFY_API_KEY in the configuration file)

Solution:

1. Confirm Dify service is started and running normally
2. Check DIFY_BASE_URL configuration (e.g., <http://localhost/v1>)
3. Verify DIFY_API_KEY is correct
4. Use curl to test connection: `curl -H "Authorization: Bearer {API_KEY}" {BASE_URL}/meta`

Error 1.6.2: Model Request Failed

Error Log:

```
The model request failed. Check whether the dify or AI model is normal.
Error Log: {error_message}
```

Cause:

- Dify service exception
- Model service down
- Model api has no quota
- Timeout

Solution:

1. Check Dify service status
 2. View Dify service logs
 3. Check api account
 4. Restart Dify service
-

1.7 Motion Control Related Errors

Error 1.7.1: TF Transform Acquisition Failed

Error Log:

```
Failed to get base_link-roi transform after {max_attempts} attempts, check if tf_tree is normal ?
```

Cause:

- TF tree incomplete
- pcl_segment node not publishing TF
- Object tracking lost the target

Solution:

1. Use `ros2 run tf2_tools view_frames` to view TF tree
 2. Confirm pcl_segment node is running normally
 3. Ensure the node is tracking the target
-

Error 1.7.2: TF Timestamp Too Old

Warning Log:

```
tf too old for base_link-roi: {age:.3f}s ago (attempt {attempt}/{max_attempts})
```

Cause:

- TF publishing frequency too low
- System delay too high, CPU load too high
- Object tracking lost the target

Solution:

1. Increase TF publishing frequency
 2. Optimize system performance
 3. Increase `tf_tolerance` parameter value
 4. Close nodes that are not needed
-

Error 1.7.3: Joint Values Not Initialized

Warning Log:

```
Joint1, joint_2, joint_4 values not initialized, skipping gaze control.
```

Cause:

- Robotic arm query service call failed
- Initialization process not completed

Solution:

1. Check if the robotic arm service is working normally
2. Verify if the arm_util node is started

1.8 Audio Playback Related Errors

Error 1.8.1: Wake-up Audio Load Failed

Error Log:

```
Failed to load the wake-up word audio file :{e}
```

Cause:

- Audio file directory does not exist
- File format not supported

Solution:

1. Check `system_vioce/zh/wakeup` or `system_vioce/en/wakeup` directory
 2. Confirm WAV file exists and format is correct
 3. Check file read permissions
 4. Reinstall the multi_brains_pre package
-

Section 2: waste_classify Module

2.1 Model Loading Related Errors

Error 2.1.1: Model File Not Found

Error Log:

```
Model file not found: {model_path}
```

Cause:

- YOLO model path configuration error
- File deleted or moved

Solution:

1. Check `model_path` parameter configuration
 2. Confirm model file exists
 3. Re-download the YOLO model file from the attached source code
-

Error 2.1.2: YOLO Model Initialization Error

Error Log:

```
YOLO Model initialization error :{e}
```

Cause:

- Factory system python environment corrupted
- Model file format error

- Insufficient memory

Solution:

1. Check if the model file is a valid ONNX format
 2. Try using CPU mode: set `device` parameter to "cpu"
 3. Check if system memory is sufficient
-

Error 2.1.3: Model Inference Start Failed

Error Log:

```
Model inference failed to start: {str(e)}
```

Cause:

- Shared memory creation failed
- Multi-process start failed
- Insufficient resources

Solution:

1. Check system shared memory limit: `sysctl kernel.shmmax`
 2. View system logs for detailed error information
-

2.2 Image Detection Related Errors

Error 2.2.1: Image Shape Not Initialized

Error Log:

```
Image shape not initialized, please ensure the camera is running and publishing images
```

Cause:

- Camera not started
- Image topic not publishing

Solution:

1. Confirm camera node is started
 2. Check if topic `/camera/color/image_raw` has data: `ros2 topic echo /camera/color/image_raw`
 3. Restart the carbase launch file
-

Error 2.2.2: Image Save Failed

Error Log:

```
Failed to save image: {e}
```

Cause:

- Output directory does not exist
- Insufficient disk space
- Camera node not started

Solution:

1. Check file path configuration

Section 3: pcl_seg Point Cloud Segmentation Module

3.1 Depth Image Processing Errors

Error 3.1.1: Invalid ROI Region (Common)

Warning Log:

```
Invalid ROI Region: [umin vmin umax vmax]
```

Cause:

- Tracking box coordinates exceed image boundaries
- Tracking box size is zero or negative
- Target area has no depth information

Solution:

1. Check if tracking box coordinates are reasonable
2. Use rviz to check if the target has depth information

Error 3.1.2: No Depth Image

Warning Log:

```
No depth image received.
```

Cause:

- Depth camera not started
- Topic subscription failed
- Depth image not publishing

Solution:

1. Confirm depth camera driver is started
 2. Check if topic `/camera/depth/image_raw` has data
 3. Verify camera hardware connection
 4. Restart the camera node
-

Error 3.1.3: No Valid Depth Points in Circular Region

Warning Log:

No valid depth points found in the circular region.

Cause:

- Target point too far, too close, or no depth information near the target point
- Depth value invalid (0 or exceeding threshold)

Solution:

1. Adjust camera observation angle or target point position
 2. Check if the depth camera is working normally
 3. Use rviz to check if the target has depth information
 4. Reduce ambient lighting
-

Error 3.1.4: Target Point Outside Image Boundary

Warning Log:

The target point is outside the image boundary. Image size: {width}x{height}, Requested: ({x}, {y})

Cause:

- Requested coordinates exceed image range
- Coordinate calculation error
- Image resolution changed

Solution:

1. Verify coordinate calculation logic
 2. Check image resolution configuration
 3. Increase boundary checking
 4. Use valid coordinate values
-

Error 3.1.5: Invalid Radius Value

Warning Log:

Invalid radius value.

Cause:

- Radius parameter is negative or zero
- Parameter passing error

Solution:

1. Ensure radius value is positive
 2. Check service call parameters
 3. Use reasonable radius values (recommended 5-50 pixels)
-

3.2 TF Transform Related Errors

Error 3.2.1: TF Transform Not Available

Warning Log:

```
TF transform not available between base_link and camera_color_optical_frame.
```

Error Log:

```
Failed to lookup transform: {ex.what()}
```

Cause:

- Missing necessary transforms in TF tree
- Camera coordinate system not configured
- TF publishing node not started

Solution:

1. Use `ros2 run tf2_tools view_frames` to check TF tree
 2. Ensure camera driver publishes correct TF
 3. Check if the transform from `base_link` to camera is correctly defined
-

Error 3.2.2: TF Transform Failed

Error Log:

```
Failed to transform actual_pose: {ex.what()}  
Failed to transform estimate_pose: {ex.what()}  
Failed to transform point: {ex.what()}
```

Cause:

- Coordinate frame does not exist
- Transform data abnormal
- Timestamp issue

Solution:

1. Check if coordinate frame names are correct
 2. Verify TF data validity
 3. Check timestamp synchronization
 4. Use `ros2 run tf2_tools echo` to view TF data
-

3.3 Point Cloud Processing Related Warnings

Error 3.3.1: ROI Point Cloud Empty

Debug Log:

```
ROI cloud is empty, skipping this depthframe
```

Cause:

- No valid depth values in ROI area
- Distance threshold set too small
- Target exceeds distance range

Solution:

1. Increase `distance_threshold` parameter
 2. Adjust ROI area size
 3. Ensure target is within effective distance
 4. Check depth image quality
-

Error 3.3.2: No Cluster Found**Debug Log:**

```
No cluster found  
No suitable cluster found
```

Cause:

- Point cloud density insufficient
- Clustering parameter settings improper
- Target too small or too far

Solution:

1. Decrease `cluster_min_size` parameter
 2. Increase `cluster_max_size` parameter
 3. Adjust `clusterTolerance` parameter
 4. Ensure target is close enough and clear
-

Error 3.3.3: No Valid Center Data**Warning Log:**

```
GetBoxCenter: no valid data.
```

Cause:

- No depth image processed yet
- Clustering failed

Solution:

1. Ensure depth image is being processed normally
 2. Check if point cloud segmentation succeeded
-

Section 4: super_tracker Object Tracking Module

4.1 Tracker Initialization Errors

Error 4.1.1: Tracker Creation Failed

Error Log:

```
Failed to create tracker: {e.what()}
```

Cause:

- Model file does not exist

Solution:

1. Check model file path: `$HOME/MODELS/tracker/mixformer_v2_sim.onnx`
 2. Adjust `num_threads` parameter
-

Error 4.1.2: Display Window Configuration Conflict

Error Log:

```
display_windows and display_windows_only_track cannot both be true. Setting both to false.
```

Cause:

- Both display parameters set to true simultaneously
- Configuration logic conflict

Solution:

1. Enable only one of the parameters:
 - `display_windows`: Always display window, support mouse selection
 - `display_windows_only_track`: Display window only during tracking
 2. Modify the configuration file to ensure they are not both true
-

4.2 Image Frame Processing Warnings

Error 4.2.1: Empty or Invalid Frame Received

Warning Log:

```
Received empty or invalid frame from camera
```

Cause:

- Camera not properly publishing images

Solution:

1. Check camera node running status
 2. Use `ros2 topic hz /camera/color/image_raw` to check publishing frequency
-

4.3 ROI Selection Errors

Error 4.3.1: Selecting ROI on Empty Frame

Error Log:

```
Cannot select ROI on empty frame
```

Cause:

- No image frame received yet
- Image frame is empty

Solution:

1. Wait for the camera to properly publish images before selecting ROI
 2. Check camera connection
 3. Confirm image topic has data
-

Error 4.3.2: Invalid ROI Selected

Warning Log:

```
Invalid ROI selected
```

Cause:

- User cancelled ROI selection
- Selected ROI width/height is zero or negative

Solution:

1. Re-select a valid ROI area
 2. Ensure the ROI area has sufficient size (recommended at least 20x20 pixels)
 3. Completely frame the target on the image
-

Error 4.3.3: Setting Track ROI Failed

Error Log:

```
Invalid frame when setting track ROI  
Failed to Set Track ROI {e.what()}
```

Cause:

- Image frame is empty
- Tracker initialization exception
- ROI coordinates invalid

Solution:

1. Ensure a valid image frame is passed in
2. Check if ROI coordinates are within the image range
3. Verify tracker status
4. Re-initialize the tracker

Error 4.3.4: Tracking Start Failed

Error Log:

```
Failed to start tracking
```

Cause:

- Current camera image invalid
- ROI setting failed
- Tracker exception

Solution:

1. Confirm camera image is normal
2. Check ROI coordinate validity
3. View specific error logs

Section 5: General Troubleshooting Suggestions

5.1 Log Level Adjustment

- If you encounter problems, enable debug log mode and send the complete error content to technical support
- ROS log level: `ros2 launch xxxx --ros-args --log-level debug`

5.2 Common Diagnostic Commands

```
# View topic list
ros2 topic list

# View topic data
ros2 topic echo <topic_name>

# View service list
ros2 service list

# View Action list
ros2 action list

# View TF tree
ros2 run tf2_tools view_frames

# Check node status
ros2 node list

# View logs
ros2 launch xxxx --ros-args --log-level debug
```

